

Module 05: Action — Expected Free Energy

Course: Active Inference 101 | **Unit:** Mathematical Frameworks | **Audience:**
First-semester undergraduates

Learning Objectives

1. Define **Expected Free Energy (EFE)** $G(\pi)$ as the criterion for policy selection.
2. Show how EFE decomposes into **pragmatic value** (goal-seeking) and **epistemic value** (information-seeking).
3. Explain how this decomposition naturally balances **exploitation and exploration**.

Introduction

Modules 01-04 covered how agents infer hidden states. But agents also need to *act*. How does the math handle action selection? The answer is **Expected Free Energy** — a quantity the agent evaluates for each possible policy to decide what to do.

Key Concepts

1. Expected Free Energy — Definition

Expected Free Energy $G(\pi)$ evaluates a policy π by asking: "If I follow this policy, how much free energy will I expect in the future?"

$$G(\pi) = \mathbb{E}_{\tilde{q}}[\log \tilde{q}(s_{\tau}) - \log P(o_{\tau}, s_{\tau})]$$

Where $\tilde{q}$ denotes the posterior predictive distribution — what the agent predicts will happen *if* it follows policy π .

Lower $G(\pi)$ = better policy. The agent selects policies with the lowest expected free energy.

2. Policy Selection via Softmax

Policies are selected probabilistically using a softmax function:

$$\mathbf{P}(\pi) = \sigma(-\gamma \times \mathbf{G}(\pi))$$

Where:

- σ : The softmax function — converts scores into probabilities
- γ : The precision parameter for policy selection — higher γ means more decisive action selection

3. The Key Decomposition — Pragmatic + Epistemic Value

EFE decomposes into two terms:

$$\mathbf{G}(\pi) \approx -\mathbb{E}_{\tilde{\mathbf{q}}}[\log \mathbf{P}(\mathbf{o}_{\tau})] + \mathbb{E}_{\tilde{\mathbf{q}}}[\mathbf{H}[\mathbf{P}(\mathbf{o}_{\tau} | \mathbf{s}_{\tau})]]$$

Or equivalently: $\mathbf{G}(\pi) = \text{Pragmatic Value} - \text{Epistemic Value}$

- **Pragmatic value** = $-\mathbb{E}_{\tilde{\mathbf{q}}}[\log \mathbf{P}(\mathbf{o}_{\tau})]$: Do expected observations match preferences (the C vector)? High when future observations are preferred.
- **Epistemic value** = $-\mathbb{E}_{\tilde{\mathbf{q}}}[\mathbf{H}[\mathbf{P}(\mathbf{o}_{\tau} | \mathbf{s}_{\tau})]]$: How much will I learn? High when the policy leads to informative observations that reduce uncertainty.

4. Pragmatic Value — Goal-Seeking (Exploitation)

Pragmatic value captures "Will this policy help me get what I want?":

$$\text{Pragmatic value} = \mathbb{E}_{\tilde{\mathbf{q}}}[\log \mathbf{P}(\mathbf{o}_{\tau})]$$

If the expected observations under policy π match the preferred observations C, pragmatic value is high. This drives **exploitation** — taking actions to achieve goals.

5. Epistemic Value — Information-Seeking (Exploration)

Epistemic value captures "Will this policy help me learn about the world?":

$$\text{Epistemic value} = -\mathbb{E}_{\tilde{\mathbf{q}}}[\mathbf{H}[\mathbf{P}(\mathbf{o}_{\tau} | \mathbf{s}_{\tau})]] \approx \mathbf{I}[\mathbf{s}_{\tau}; \mathbf{o}_{\tau} | \pi]$$

This measures how much mutual information there is between future states and observations under policy π . High epistemic value means the policy reduces uncertainty about hidden states. This drives **exploration** — seeking information.

6. The Exploration-Exploitation Balance

EFE automatically balances exploration and exploitation:

- **When uncertainty is high:** Epistemic value dominates $\rightarrow$ the agent explores (seeks information)
- **When uncertainty is low:** Pragmatic value dominates $\rightarrow$ the agent exploits (pursues goals)

This is not a parameter you tune — it emerges naturally from the math.

Summary

Expected Free Energy $G(\pi)$ is the criterion for selecting policies in Active Inference. It decomposes into pragmatic value (goal-seeking/exploitation) and epistemic value (information-seeking/exploration). The softmax function converts EFE scores into policy probabilities, with precision γ controlling decisiveness. The exploration-exploitation balance emerges naturally from the mathematics.

Further Reading

- Friston, K. J. et al. (2015). Active inference and epistemic value. *Cognitive Neuroscience*, 6(4), 187-214.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press. (Chapter 6)
- Sajid, N. et al. (2021). Active inference: Demystified and compared. *Neural Computation*, 33(3), 674-712.